

Personal Finance Tracker - Project Report

1. Introduction

The Personal Finance Tracker is a console-based application developed using C++.

It helps users manage their monthly income, expenses, and budget.

The system uses Object-Oriented Programming (OOP) principles such as inheritance, polymorphism, abstraction, and encapsulation.

2. Objectives

- Track income and expenses.
- Set and monitor monthly budgets.
- Generate monthly financial reports.
- Demonstrate OOP concepts in a practical project.

3. Technologies Used

- C++ Programming Language
- OOP Concepts
- Console-based user interface

4. OOP Concepts Used

- Encapsulation: Getter methods in Transaction class.
- Inheritance: Income and Expense classes inherit from Transaction.
- Polymorphism: Overridden display() and getType() methods.
- Abstraction: Account class hides implementation details.

5. System Design

The system contains four main classes:

1. Transaction (Base Abstract Class)

2. Income (Derived Class)
3. Expense (Derived Class)
4. Account (Manages all functionalities)

Additionally, a Budget class is used to manage monthly spending limits.

6. Features of the System

- Add income and expenses.
- View all transactions.
- Set monthly budget.
- Generate monthly summary.
- Automatic budget check.

7. Output Description

The program displays all recorded transactions and generates a monthly report showing:

- Total Income
- Total Expense
- Savings
- Budget status (Exceeded or within limit)

8. Conclusion

The Personal Finance Tracker successfully demonstrates how C++ and OOP principles can be used to build

a functional finance management tool. The project is user-friendly, efficient, and helps users maintain

better financial discipline.

9. Future Scope

- Add GUI interface.
- Export reports to files.
- Add database storage.

- Provide graphical analysis charts.